# 6.3: Matrices for Linear Transformations

MATH 2360

**Definitions:**

1. **Standard Matrix for a Linear Transformation Thm:** Let $\mathcal{T}: \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation such that, for the standard basis vectors, $\vec{e}_i \in \mathbb{R}^n$,

$$T(\vec{e}_1) = \begin{bmatrix} a_{11} \\ a_{21} \\ \cdots \\ a_{m1} \end{bmatrix}, T(\vec{e}_2) = \begin{bmatrix} a_{12} \\ a_{22} \\ \cdots \\ a_{m2} \end{bmatrix}, \ldots, T(\vec{e}_n) = \begin{bmatrix} a_{12} \\ a_{22} \\ \cdots \\ a_{mn} \end{bmatrix}$$

Then the $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is such that $T(\vec{v}) = A\vec{v}$ for every $\vec{v} \in \mathbb{R}^n$, and is the **standard matrix** for $\mathcal{T}$.

**Examples:**

1) Find the standard matrix for the linear transformation given below.

$$T(x, y) = (x + 2y, x - 2y)$$

**Definition:**

2. The composition $\mathcal{T}$ of $\mathcal{T}_1: \mathbb{R}^n \to \mathbb{R}^m$ with $\mathcal{T}_2: \mathbb{R}^m \to \mathbb{R}^p$ is defined

$$\mathcal{T}(\vec{v}) = \mathcal{T}_2(\mathcal{T}_1(\vec{v}))$$

where $\vec{v} \in \mathbb{R}^n$ is a vector, denoted $\mathcal{T} = \mathcal{T}_2 \circ \mathcal{T}_1$

3. **Composition of Linear Transformations Thm:** Let $\mathcal{T}_1: \mathbb{R}^n \to \mathbb{R}^m$ with $\mathcal{T}_2: \mathbb{R}^m \to \mathbb{R}^p$ is be linear transformations with standard matrices $A_1$ and $A_2$, respectively. The **composition** $\mathcal{T}: \mathbb{R}^A \to \mathbb{R}^p$ defined by $\mathcal{T}(\vec{v}) = \mathcal{T}_2(\mathcal{T}_1(\vec{v}))$ is a linear transformation, with standard matrix $A = A_2A_1$.

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**Example:**

2) Find the standard matrices $A$ and $A'$ for $\mathcal{T} = \mathcal{T}_2 \circ \mathcal{T}_1$ and $\mathcal{T}' = \mathcal{T}_1 \circ \mathcal{T}_2$ if

$\mathcal{T}_1 : \mathbb{R}^2 \to \mathbb{R}^2,\ \mathcal{T}_1(x, y) = (x - 2y, 2x + 3y)$ and $\mathcal{T}_2 : \mathbb{R}^2 \to \mathbb{R}^2,\ \mathcal{T}_2(x, y) = (y, 0)$

**Definition:**

4. If $\mathcal{T}_1 : \mathbb{R}^n \to \mathbb{R}^n$ and $\mathcal{T}_2 : \mathbb{R}^n \to \mathbb{R}^n$ are linear transformations such that for every $\vec{v} \in \mathbb{R}^n$, $\mathcal{T}_2(\mathcal{T}_1(\vec{v})) = \vec{v}$ and $\mathcal{T}_1(\mathcal{T}_2(\vec{v})) = \vec{v}$, then $\mathcal{T}_2$ is the **inverse** of $\mathcal{T}_1$ is **invertible**. The inverse is denoted $\mathcal{T}_1^{-1}$ and, if it exists, the inverse is unique.

5. **Equivalent Conditions for Invertibility Thm:** Let $\mathcal{T}$ be a linear transformation with standard matrix $A$. Then the following are equivalent:
   (1) $\mathcal{T}$ is invertible.
   (2) $\mathcal{T}$ is an isomorphism.
   (3) $A$ is invertible. (Also, the standard matrix for $\mathcal{T}^{-1}$ is $A^{-1}$.)

**Example:**

3) Show that $\mathcal{T} : \mathbb{R}^2 \to \mathbb{R}^2$, $\mathcal{T}(x, y) = (x + y, x - y)$ is invertible and find its inverse.

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**Definition:**

6. **Transformation Matrix for Nonstandard Bases**: Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with bases $B$ and $B'$, respectively, $\boldsymbol{B} = \{\vec{\boldsymbol{v}}_{\boldsymbol{1}}, \vec{\boldsymbol{v}}_{\boldsymbol{2}}, \dots, \vec{\boldsymbol{v}}_{\boldsymbol{n}}\}$**.** If $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ is a linear transformation such that

$$[\mathcal{T}(\vec{v}_1)]_{B'} = \begin{bmatrix} \boldsymbol{a_{11}} \\ \boldsymbol{a_{21}} \\ \dots \\ \boldsymbol{a_{m1}} \end{bmatrix}^{\boldsymbol{T}}, \begin{bmatrix} \boldsymbol{a_{12}} \\ \boldsymbol{a_{22}} \\ \dots \\ \boldsymbol{a_{m2}} \end{bmatrix}^{\boldsymbol{T}}, \dots, [\boldsymbol{\mathcal{T}}(\overrightarrow{v_n})]_{\boldsymbol{B'}} = \begin{bmatrix} \boldsymbol{a_{1n}} \\ \boldsymbol{a_{2n}} \\ \dots \\ \boldsymbol{a_{mn}} \end{bmatrix}^{\boldsymbol{T}}$$

Then

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Is such that $[\mathcal{T}(\vec{v})]_{B'} = A[\vec{v}]_B$ for every $\vec{v} \in \mathcal{V}$. The matrix $A$ is the **matrix of $\boldsymbol{\mathcal{T}}$ relative to the bases $\boldsymbol{B}$ and $\boldsymbol{B}'$**.

**Example:**

4) If $\mathcal{T}: \mathbb{R}^2 \to \mathbb{R}^3$ such that $\mathcal{T}(x, y) = (x + y, x - y)$, find the matrix for $\mathcal{T}$ relative to bases $B = \{(1, -1), (0, 1)\}$ and $B' = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$.

5) Using the matrix $A$ found above, find $\mathcal{T}(\vec{v})$ if $\vec{v} = (5, 4)$.

**Definitions:**

7. For the case where $\mathcal{V} = \mathcal{W}$ and $B = B'$, matrix $A$ is called the **matrix of $\mathcal{T}$ relative to the basis $\boldsymbol{B}$**, in which case the matrix of the identity transformation is $I_n$.

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